

Worksheet 6: Everyday Mathematics and Real-World Patterns

Topic Focus: Using patterns to solve word problems, calendars, and spatial puzzles.

1. In a calendar month, if the 3rd is a Tuesday, what day is the 17th?

- A) Monday B) Tuesday C) Wednesday D) Thursday

Solution: Days repeat every 7 days. $3 + 7 = 10\text{th}$ (Tues). $10 + 7 = 17\text{th}$ (Tues).

2. Look at any 3x3 square of dates on a calendar. The sum of opposite corners are always:

- A) Prime numbers B) Equal to each other C) Multiples of 10 D) Odd numbers

Hint: Try it,, If corners are (n) and $(n+8)$, sum is $2n+8$. Other corners are $(n+2)$ and $(n+6)$, sum is $2n+8$.

3. Staircase pattern: Step 1 needs 1 block. Step 2 needs 3 blocks total. Step 3 needs 6 blocks total. Blocks for 5 steps?

- A) 10 B) 12 C) 15 D) 20

Hint: Building a staircase creates triangular numbers (1, 3, 6, 10, 15). So add $+2, +3, +4, +5, \dots$

4. Rows of chairs: Row 1 = 10. Row 2 = 13. Row 3 = 16. How many in Row 6?

- A) 22 B) 25 C) 28 D) 31

Hint: Adding 3 per row. Row 4=19, Row 5=22, Row 6=25.

5. Savings: Week 1=10rs. Week 2=20rs. Week 3=40rs. Week 4=80rs. Savings in Week 6?

- A) 160rs B) 240rs C) 320rs D) 640rs

Hint: The money is doubling every week. Week 5=160, Week 6=320.

6. Pizza cuts (straight lines): 1 cut=2 pieces. 2 cuts=4 pieces max. 3 cuts=7 pieces max. 4 cuts max pieces?

- A) 8 B) 9 C) 10 D) 11

Hint: The differences added are $+2, +3, +4$. So $7 + 4 = 11$. (Central polygonal numbers).

7. 5 friends meet and everyone shakes hands exactly once. Total handshakes?

- A) 5 B) 10 C) 15 D) 20

Hint: Friend A shakes 4 hands. Friend B shakes 3 (already shook A). C shakes 2. D shakes 1. $4+3+2+1 = 10$.

8. Digital clock shows 12:00. Next identical digits is 1:11, then 2:22. How many times between 12:00 and 6:00?

- A) 4 B) 5 C) 6 D) 7

Hint: Count them: 1:11, 2:22, 3:33, 4:44, 5:55.

9. Tiling: 1x1 floor=1 tile. 2x2 floor=4 tiles. 3x3=9 tiles. How many tiles Added to change 4x4 to 5x5?

- A) 5 B) 7 C) 9 D) 11

Hint: 5×5 is 25. 4×4 is 16. $25 - 16 = 9$. Adding an outer layer requires an odd number of tiles.

10. Secret code: A=1, B=3, C=5, D=7. What number represents F?

- A) 9 B) 10 C) 11 D) 13

Hint: Letters are mapped to odd numbers. E would be 9, so F is 11.

11. A knockout tennis tournament has 32 players. How many matches must be played to find One winner?

- A) 16 B) 31 C) 32 D) 64

Hint: Every match eliminates exactly 1 player. To eliminate 31 players and leave 1 winner takes 31 matches.

12. A bus has seats numbered 1, 2, 3, 4 on the first row (two left, two right). Window seats on the left are 1, 5, 9... What is the rule?

- A) Add 3 B) Add 4 C) Multiples of 4 D) Prime numbers

Hint: Because there are 4 seats per row, the pattern jumps by 4 each time.

13. You have a 3 digit combination lock using numbers 0-9. How many possible combinations exist?

- A) 30 B) 100 C) 999 D) 1000

Hint: 10 options for slot 1 x 10 for slot 2 x 10 for slot 3. (10^3). Patterns of possibilities multiply.

14. Stacking bricks: A wall is 1 brick wide on top, 2 bricks in row below, 3 in row below. If the base has 8 bricks, how many total bricks in the wall?

- A) 24 B) 28 C) 36 D) 45

Hint: Sum of 1 through 8.

15. A grandfather clock chimes once at 1:00, twice at 2:00, etc. How many times does it chime between 12:30 and 4:30?

- A) 6 B) 10 C) 15 D) 20

Hint: Add the chimes for 1:00, 2:00, 3:00, and 4:00. $1+2+3+4 = 10$.